

Vishal Naveen

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EDUCATION

University of Florida

B.S. in Electrical Engineering, Minor in Computer Science | GPA: 4.0

- Honors Program; Florida Academic Scholars and UF Honors Scholarship.

May 2029
Gainesville, FL

EXPERIENCE

Pedro Pathing

Algorithm Developer and Beta Tester, Open-Source Robotics Library

- Implemented path following over multi-order Bézier curves in Java, tuning velocity profiling and PID heading correction.
- Tested pre-release builds of an autonomous navigation library, **used by 2,000+ FTC teams**, documenting pathing regressions.

Jun 2025 – Jan 2026

Remote

University of South Florida

Student Researcher, Coding Theory and Reinforcement Learning

- Built a deep Q-network bit-flipping decoder for LDPC codes in stable-baselines3, training the agent to select bit flips from syndrome state and benchmarking it against greedy bit-flipping and BP+LSD baselines across 270 seeded trials.
- Traced a 25.9% versus 93.7% syndrome-success gap behind greedy bit-flipping to the parity-check matrix, not the policy.
- Reduced evaluation from 103 minutes to 5.4 seconds, **a 1,100x speedup**, via model caching, feasibility gating, and parallelization.
- Open-sourced the study as the project's only pre-college researcher: 810 trial records, a 3,600-trial sweep, and verification scripts.

Dec 2024 – Jan 2026

Tampa, FL

Inc to Inc

Software Developer

- Developed 3 production websites with one other developer for STEAMA and Tampa Youth Basketball Association.
- Shipped session authentication, class scheduling, PayPal payment and donation flows, and storefronts on a relational SQL schema.

Jun 2023 – Jul 2025

Tampa, FL

PROJECTS

TactileVLA-Edge

Independent Robotics Research: LeRobot, PyTorch, ACT, SmolVLA

- Trained an ACT policy and fine-tuned SmolVLA (450M) in LeRobot on a single consumer GPU: **18/20 and 16/20 success on deformable-object pick-and-place**.
- Collected 160 teleoperated demonstrations, 20 per cell across the 8 outer cells of a 3x3 grid, sweeping 19 grasp orientations.
- Assembled an SO-101 leader-follower teleoperation rig for under \$300 and published the dataset and evaluation protocol.
- Demonstrated spatial generalization: **both policies solved held-out cell B2**, absent from all 160 training episodes.
- Submitted a patch to Hugging Face LeRobot (PR #3798) diagnosing a streaming-dataset collate crash on unconverted PIL images: 86 lines across 2 files with a regression test.

Jun 2026 – Present

Sixth Sense

Independent iOS Project: Swift, Core ML, YOLOv8, ARKit, OpenCV

- Built an assistive navigation iOS app for blind and low-vision users, **running a 12.7 MB YOLOv8n Core ML detector at a sustained 15 fps on iPhone 14 Pro**.
- Modeled obstacle distance with no depth sensor, applying monocular pinhole geometry to a 21-class reference-size table and Kalman-filtering the mean of width-, height-, and area-derived estimates.
- Recorded indoor routes with 6-DoF ARKit tracking and scoped accessibility requirements with the City of Tampa ADA coordinator.

Sep 2023 – Jun 2025

TECHNICAL SKILLS

Languages: Python, Java, C++, Swift, JavaScript, SQL

Machine Learning: PyTorch, LeRobot, stable-baselines3, Hugging Face, NumPy, OpenCV, YOLOv8, Core ML, ACT, SmolVLA

Robotics: imitation learning, reinforcement learning (DQN), PID control, Bézier path planning, teleoperation, CAD

Tools: Git, GitHub, Linux, Jupyter, Xcode, Android Studio, React, Node.js, REST APIs, ARKit

LEADERSHIP AND INVOLVEMENT

Middleton Robotics

FTC Team Captain and Lead Programmer, Teams 4997 and 26588

- Led a 15-member team to its first state title in 7 years and programmed a world-record-tying 105-point autonomous routine.
- Raised \$25,000 in team funding and organized 15 outreach events and technical workshops.
- Owned the team codebase on GitHub and ran build-season sprint planning across the mechanical and software subteams.

Aug 2023 – May 2026

Tampa, FL

Achievements: FTC Innovate Award 1st Place 2026; FTC Inspire Award 2nd Place 2025; FTC State Finalist and Alliance Captain 2025; FRC Rising All-Star Award, FIRST World Championship 2025; Congressional App Challenge 3rd Place 2024

Activities: Co-founded a VEX robotics camp teaching C++ to 12 under-resourced middle schoolers who formed 2 competitive teams, 2024–2025; programmed for FRC Team 1369 through a FIRST World Championship season; Mayor's Youth Corps, 1 of 40 citywide